Vibrations Meeting

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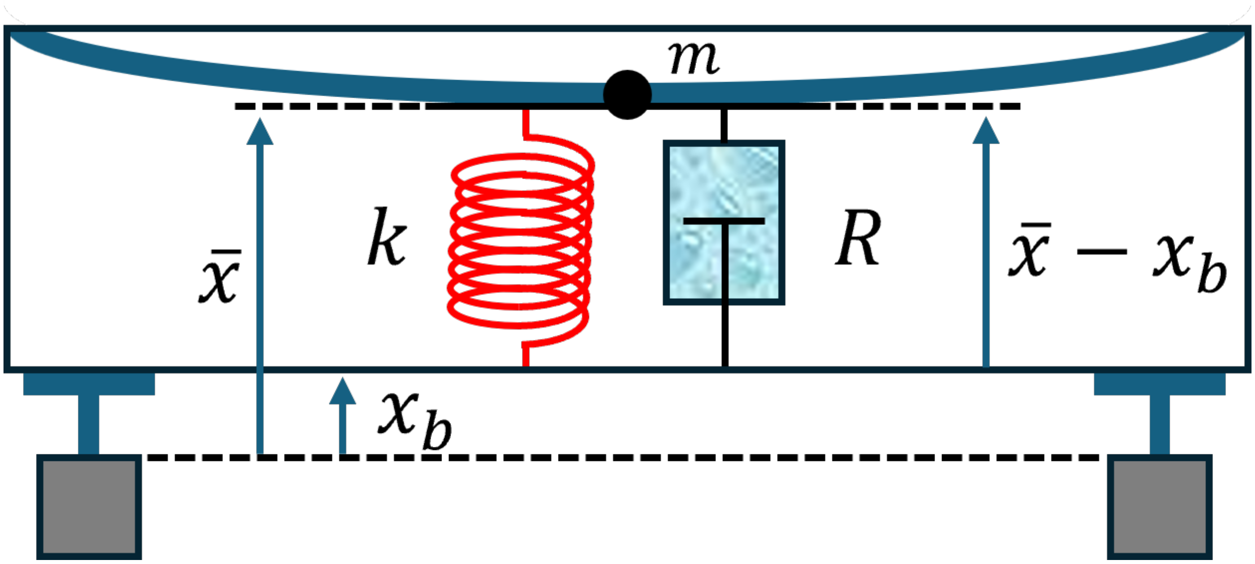

Figure 1: Physical Sketch

We model the vibrating beam supported at two ends as a base excitation problem. Upon concentrating all of the mass of the beam at its center, the lagrangian of the system can be given as such:

$$\mathcal{L} = \frac{1}{2}m\dot{\bar{x}}^2 - \frac{1}{2}k(\bar{x} - x_b)^2$$

Likewise, the non-conservative dissipative function, which accounts for the damping of the mass m, can be given as such:

$$D = -\frac{1}{2}R(\dot{\bar{x}} - \dot{x}_b)^2$$

The EOM of the system is given as such (in accordance with Newton's 2nd Law):

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\bar{x}}} - \frac{\partial \mathcal{L}}{\partial \bar{x}} = \frac{\partial D}{\partial \dot{\bar{x}}} \implies m\ddot{\bar{x}} + k(\bar{x} - x_b) = -R(\dot{\bar{x}} - \dot{x}_b) \implies \boxed{m\ddot{\bar{x}} + R\dot{\bar{x}} + k\bar{x} = R\dot{x}_b + kx_b}$$

Upon defining $\omega_n = \sqrt{k/m}$ and $\zeta = R/2m\omega_n$, this can be equivalently formulated as follows:

$$\boxed{\ddot{\bar{x}} + 2\zeta\omega_n\dot{\bar{x}} + \omega_n^2\bar{x} = 2\zeta\omega_n\dot{x}_b + \omega_n^2 x_b}$$

We can now identify the forcing function (per unit mass) of the system:

$$f(t) = 2\zeta\omega_n\dot{x}_b(t) + \omega_n^2 x_b(t)$$

With the understanding that the base displacement function can be expressed as a sinusoid that *turns on* at $t = 0$, we may say for some A and ω_f (with $u(t)$ being the unit step function):

$$x_b(t) = A\sin(\omega_f t)\,u(t) \implies f(t) = A(2\zeta\omega_n\omega_f\cos(\omega_f t) + \omega_n^2\sin(\omega_f t))\,u(t)$$

Now let $\beta = \zeta\omega_n$ and $\omega_d = \omega_n\sqrt{1-\zeta^2}$. It thus follows that the EOM and the forcing function (per unit mass) can be expressed as follows:

$$\boxed{\ddot{\bar{x}} + 2\beta\dot{\bar{x}} + \omega_n^2\bar{x} = f(t) \quad \text{with} \quad f(t) = A(2\beta\omega_f\cos(\omega_f t) + \omega_n^2\sin(\omega_f t))\,u(t)}$$

The Green's Function for this system has the following form:

$$G(t) = \frac{1}{\omega_d}e^{-\beta t}\sin(\omega_d t)u(t)$$

Upon combining the general homogeneous solution to the system x_h (i.e. $A = 0$) to the particular solution generated by the Green's Function, the general solution to the forced system can be obtained as follows:

$$\begin{aligned}
\bar{x}(t) &= \bar{x}_h(t) + \int_{-\infty}^{\infty} f(\tau)\,G(t-\tau)\,d\tau \\
&= \bar{x}_h(t) + \int_{-\infty}^{\infty} A(2\beta\omega_f\cos(\omega_f\tau) + \omega_n^2\sin(\omega_f\tau))\,u(\tau)\,\frac{1}{\omega_d}e^{-\beta(t-\tau)}\sin(\omega_d(t-\tau))u(t-\tau)\,d\tau \\
&= \bar{x}_h(t) + \frac{A}{\omega_d}\int_0^t (2\beta\omega_f\cos(\omega_f\tau) + \omega_n^2\sin(\omega_f\tau))\,e^{-\beta(t-\tau)}\sin(\omega_d(t-\tau))\,d\tau \\
&= e^{-\beta t}\left(\bar{x}_0\cos(\omega_d t) + \frac{\dot{\bar{x}}_0 + \beta\bar{x}_0}{\omega_d}\sin(\omega_d t)\right) \\
&\quad + \frac{A}{\omega_d}\int_0^t (2\beta\omega_f\cos(\omega_f\tau) + \omega_n^2\sin(\omega_f\tau))\,e^{-\beta(t-\tau)}\sin(\omega_d(t-\tau))\,d\tau
\end{aligned}$$

Therefore, the general solution to the EOM follows as such:

$$\begin{aligned}
\bar{x}(t) = e^{-\beta t}&\left(\bar{x}_0\cos(\omega_d t) + \frac{\dot{\bar{x}}_0 + \beta\bar{x}_0}{\omega_d}\sin(\omega_d t)\right) \\
&+ \frac{A}{\omega_d}\int_0^t e^{-\beta(t-\tau)}(2\beta\omega_f\cos(\omega_f\tau) + \omega_n^2\sin(\omega_f\tau))\sin(\omega_d(t-\tau))\,d\tau
\end{aligned}$$

Now, consider the EOM in the frequency domain with $\hat{\bar{x}} = \mathscr{F}\{\bar{x}\}$ and $\hat{x}_b = \mathscr{F}\{x_b\}$:

$$(-\omega^2 + 2\zeta\omega_n\omega i + \omega_n^2)\,\hat{\bar{x}}(\omega) = (2\zeta\omega_n\omega i + \omega_n^2)\,\hat{x}_b(\omega)$$

The transfer function pertaining to this system thus has the following form:

$$T(\omega) = \frac{\hat{\bar{x}}(\omega)}{\hat{x}_b(\omega)} = \frac{1 + 2\zeta\left(\frac{\omega}{\omega_n}\right)i}{1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2\zeta\left(\frac{\omega}{\omega_n}\right)i} \implies |T(\omega)| = \sqrt{\frac{1 + 4\zeta^2\left(\frac{\omega}{\omega_n}\right)^2}{\left(1 - \left(\frac{\omega}{\omega_n}\right)^2\right)^2 + 4\zeta^2\left(\frac{\omega}{\omega_n}\right)^2}}$$

Define the auxiliary function:

$$y(x) := \left|T\left(\omega_n\sqrt{x}\right)\right|^2 = \frac{1+ax}{(1-x)^2+ax} \quad \text{with} \quad a = 4\zeta^2$$

It follows that:

$$y'(x) = \frac{-ax^2-2x+2}{((1-x)^2+ax)^2} \implies \left(y'(x_{\max}) = 0 \implies -ax_{\max}^2 - 2x_{\max} + 2 = 0\right)$$

Furthermore, the (local) maximum value of y occurs when $y' = 0$ and $y'' < 0$. This occurs specifically when:

$$x_{\max} = \frac{\sqrt{1+2a}-1}{a} \implies y_{\max} = y(x_{\max}) = \frac{(\sqrt{1+2a}+1)^2}{(\sqrt{1+2a}-1)(\sqrt{1+2a}+3)}$$

It holds that $y_{\max} = |T|^2_{\max}$. If we define $M := \sqrt{y_{\max}}$, which corresponds to the* (maximum) peak of the modulus $|T(\omega)|$, then it follows that:

$$M^2 = \frac{(\sqrt{1+2a}+1)^2}{(\sqrt{1+2a}-1)(\sqrt{1+2a}+3)} \implies a = \frac{2M}{\sqrt{M^2-1}}\left(\frac{M}{\sqrt{M^2-1}}-1\right)$$

Upon recalling that $a = 4\zeta^2$, we can finally obtain:

$$\boxed{\zeta = \sqrt{\frac{M}{2\sqrt{M^2-1}}\left(\frac{M}{\sqrt{M^2-1}}-1\right)}}$$

The Damping Ratio

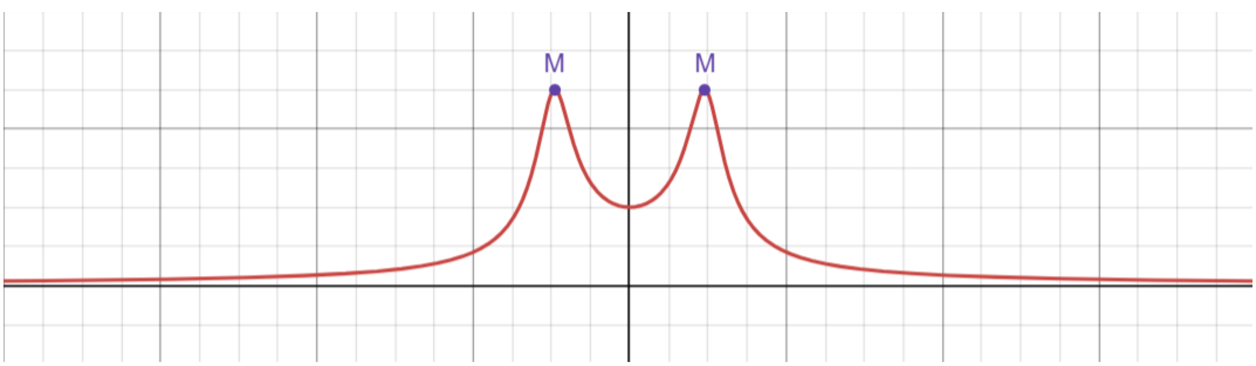

Figure 2: Peak Identification(s) of $|T(\omega)|$

*Note that there are two peaks corresponding to $\omega_{\max} = \pm\omega_n\sqrt{x_{\max}}$.

DISCLAIMER: The above results have not been checked over or scrutinized. Further analysis needs to be preformed in order to verify these results.

An alternative base excitation function may instead have the following representation:

$$x_b(t) = Au(t) \implies f(t) = A\beta\dot{u}(t) + A\omega_n^2 u(t) = A(\beta\delta(t) + \omega_n^2 u(t))$$

Just as before, using the Green's Function, a general solution can be obtained as follows:

$$\begin{aligned}
\bar{x}(t) &= \bar{x}_h(t) + \int_{-\infty}^{\infty} f(\tau)\, G(t-\tau)\, d\tau \\
&= \bar{x}_h(t) + \int_{-\infty}^{\infty} A(\beta\delta(\tau) + \omega_n^2 u(\tau))\, \frac{1}{\omega_d} e^{-\beta(t-\tau)} \sin(\omega_d(t-\tau)) u(t-\tau)\, d\tau \\
&= \bar{x}_h(t) + \frac{A}{\omega_d} \int_0^t (\beta\delta(\tau) + \omega_n^2 u(\tau)) e^{-\beta(t-\tau)} \sin(\omega_d(t-\tau))\, d\tau \\
&= \bar{x}_h(t) + \frac{A}{\omega_d} \left[\beta e^{-\beta t} \sin(\omega_d t) + \frac{\omega_n^2}{\beta^2 + \omega_d^2} \left(\omega_d - e^{-\beta t} \left(\beta \sin(\omega_d t) + \omega_d \cos(\omega_d t) \right) \right) \right]
\end{aligned}$$

$$\boxed{\bar{x}(t) = A + e^{-\beta t} \left((\bar{x}_0 - A)\cos(\omega_d t) + \frac{\dot{\bar{x}}_0 + \beta(\bar{x}_0 - A)}{\omega_d} \sin(\omega_d t) \right) \quad \text{for} \quad t > 0}$$

General Solution for $x_b(t) = Au(t)$ $(t > 0)$